

Unit 9 Progress Check: FRQ

1. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions related to H_2O .

Substance	ΔG°_f at 298 K (kJ/mol)
$\text{H}_2\text{O}(l)$	-237.2
$\text{H}_2\text{O}(g)$	-228.4

- (a) Using the information in the table above, determine the value of ΔG° at 298 K for the process represented by the equation $\text{H}_2\text{O}(l) \rightleftharpoons \text{H}_2\text{O}(g)$.



Please respond on separate paper, following directions from your teacher.

- (b) Considering your answer to part (a), indicate whether the process is thermodynamically favorable at 298 K. Justify your answer.



Please respond on separate paper, following directions from your teacher.

- (c) Considering your answer to part (b), explain why $\text{H}_2\text{O}(l)$ has a measurable equilibrium vapor pressure at 298 K.



Please respond on separate paper, following directions from your teacher.

Water vapor can be produced in two different processes, as represented below.

Process 1	Process 2
$\text{H}_2\text{O}(s) \rightleftharpoons \text{H}_2\text{O}(g)$	$\text{H}_2\text{O}(l) \rightleftharpoons \text{H}_2\text{O}(g)$

- (d) In terms of concepts of entropy and the particle-level structure of the different phases of water, explain why the change in entropy, ΔS , is greater for process 1 than for process 2.



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Please respond on separate paper, following directions from your teacher.

Part (a)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response includes both the following criteria:

- ☐ $\Delta G^\circ = \Sigma \Delta G_f^\circ \text{ products} - \Sigma \Delta G_f^\circ \text{ reactants}$
- ☐ $\Delta G^\circ = -228.4 - (-237.2) = +8.8 \text{ kJ/mol}_{rxn}$

Part (b)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response. The answer to part (b) must be consistent with part (a). If a student makes an error in a previous part of the question, the student can still earn credit in a later part if the incorrect answer is used correctly.



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The response indicates that the process is not thermodynamically favorable at **298 K** because ΔG° is positive. (OR, if part (a) indicates a negative value of ΔG° , the response indicates that the process is thermodynamically favorable because ΔG° is negative.)

Part (c)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



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0	1
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The response indicates that even though the process is not thermodynamically favorable, it does not follow that the process does not occur at all. At equilibrium, the reactant (liquid water) is avored over the product (water vapor). This does not mean that some of the liquid water cannot evaporate to become water vapor.

Part (d)

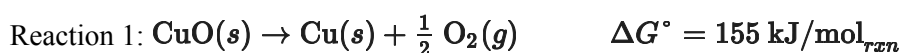
Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response indicates that the change in entropy between ice, $\text{H}_2\text{O}(s)$, and water vapor, $\text{H}_2\text{O}(g)$, is greater because the entropy of water in the solid phase, with molecules bound in fixed positions within the solid, is less than the entropy of water in the liquid phase, with molecules able to move more freely and form a greater number of spatial configurations. Water in the gas phase, with molecules much more dispersed and even more free to move, has the highest entropy of the three phases. Therefore, process 1 (sublimation) involves a greater change in entropy than process 2 (evaporation from a liquid) does.

2. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



A chemist wants to produce $\text{Cu}(s)$ from a sample of pure $\text{CuO}(s)$ according to reaction 1, represented by the equation above.

(a) Using the data in the following table, calculate the value of the standard entropy change, ΔS° , for the reaction.



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Substance	Absolute Entropy at 298 K ($\text{J K}^{-1} \text{mol}^{-1}$)
$\text{Cu}(s)$	33
$\text{O}_2(g)$	205
$\text{CuO}(s)$	43



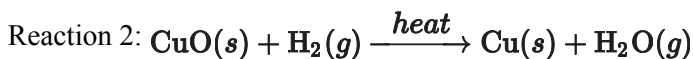
Please respond on separate paper, following directions from your teacher.

(b) Given that ΔG° for reaction 1 is positive ($155 \text{ kJ/mol}_{\text{rxn}}$), what must be true about the sign of ΔH° for the reaction? Justify your answer.



Please respond on separate paper, following directions from your teacher.

The decomposition of $\text{CuO}(s)$ into $\text{Cu}(s)$ and $\text{O}_2(g)$ is not a thermodynamically favored reaction. However, to produce $\text{Cu}(s)$ from $\text{CuO}(s)$, the chemist decides to pass $\text{H}_2(g)$ gas over the $\text{CuO}(s)$ as it is heated strongly, as represented below.

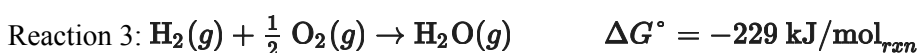


(c) Reaction 2 is what type of chemical reaction? Justify your answer.



Please respond on separate paper, following directions from your teacher.

A third chemical reaction is represented below.



(d) Show how a combination of reaction 1 and reaction 3 can be used to produce reaction 2.



Please respond on separate paper, following directions from your teacher.

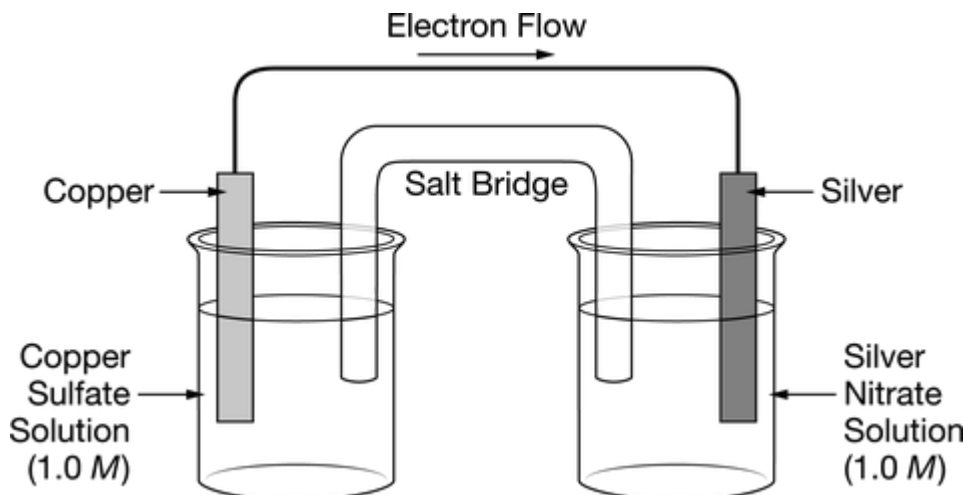
(e) Determine the value of ΔG° for reaction 2.

The chemist takes the $\text{Cu}(s)$ produced from reaction 2 and uses it to make an electrode in a galvanic



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cell, as shown in the following diagram.



Please respond on separate paper, following directions from your teacher.

(f) Using the reduction half-reactions in the table below, write a balanced net-ionic equation representing the overall reaction that takes place as the cell operates.

Half-Reaction	E° (volts)
$\text{Cu}^{2+}(\text{aq}) + 2 e^- \rightarrow \text{Cu}(\text{s})$	0.337
$\text{Ag}^+(\text{aq}) + e^- \rightarrow \text{Ag}(\text{s})$	0.800



Please respond on separate paper, following directions from your teacher.

(g) Determine the value of E° for the cell.



Please respond on separate paper, following directions from your teacher.

(h) Determine the value of ΔG° for the cell reaction.



Please respond on separate paper, following directions from your teacher.



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Part (a)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response includes the both the following set up and calculation:

$$\Delta S^{\circ} = \Sigma S^{\circ} \text{ products} - \Sigma S^{\circ} \text{ reactants}$$

$$\Delta S^{\circ} = [33 + \frac{1}{2}(205)] - 43 = +92 \text{ J}/(\text{K mol}_{rxn})$$

Part (b)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response indicates that the sign of ΔH° for the reaction must be positive because $\Delta G^{\circ} = \Delta H^{\circ} - T\Delta S^{\circ}$ implies that $\Delta H^{\circ} = \Delta G^{\circ} + T\Delta S^{\circ}$. In other words, if ΔG° and $T\Delta S^{\circ}$ are positive, then ΔH° must also be positive.

(Note: The calculation $\Delta H^{\circ} = 155 + 298(0.092) > 0$ is not required to earn the point.)

Part (c)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The response meets both of the following criteria:



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- ☐ The response indicates that the reaction is a redox (or oxidation/reduction) reaction.
- ☐ The response indicates that **Cu** is reduced (oxidation state changes from +2 to 0) and **H** is oxidized (oxidation state changes from 0 to +1).

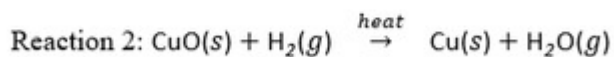
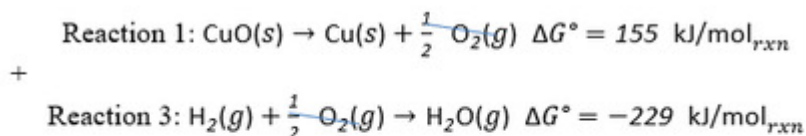
Part (d)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response shows that reaction 2 is the sum of reaction 1 and reaction 3.



Part (e)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response indicates that

$$\Delta G^\circ_{\text{reaction 2}} = \Delta G^\circ_{\text{reaction 1}} + \Delta G^\circ_{\text{reaction 3}} = 155 + (-229) = -74 \text{ kJ/mol}_{rxn}$$



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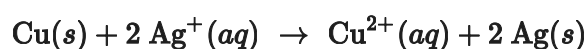
Part (f)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response. Note: phases after species are not necessary for full credit.



0	1
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The response indicates that the equation is

**Part (g)**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The response includes the following correct calculation of E° for the cell: $E^\circ = 0.800 \text{ V} - 0.337 \text{ V} = 0.463 \text{ V}$.

Part (h)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response includes both the following set up and calculation:

$$\Delta G^\circ = -nFE^\circ = -(2 \text{ mol electrons} \times \frac{96,500 \text{ C}}{1 \text{ mole electrons}} \times 0.463 \frac{\text{J}}{\text{C}}) = -89,400 \text{ J/mol}_{\text{rxn}}$$